1.2 Finding the Least Common Multiple

Lesson Introduction

Benchmark	MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.	Learning Target	• I can find the least common multiple (LCM) in mathematical and real-world contexts.

	Building On	Working Toward
Benchmark Coherence	MA.4.AR.3.1 Determine factor pairs for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither.	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

Essential Questions	• What is the least common multiple (LCM)? • What strategies can be used to determine the LCM of two whole numbers? • How does the LCM appear in real-world contexts? • What are the similarities and differences between finding the LCM and the GCF?
Lesson Narrative	This lesson introduces finding the least common multiple (LCM) of two whole numbers. The Warm-Up includes a video where students watch 3 sand timers, which then leads into the Guided Instruction around that scenario to build conceptual understanding of LCM in real-world contexts. Number line models are used to introduce multiples, common multiples, and the least common multiple. Then students are given an opportunity to practice finding the LCM to solve both real-world and mathematical problems.

Component Summary	1.2.1 Warm-Up (~5 min.)	Hourglass timing video and discussion *(SE p. 9)*	1.2.4 Collaboration (10 min.)	Open-ended task using digits 1-9 to create factors and LCM (can be completed on different day) *(SE p. 12)*
	1.2.2 Guided Instruction (10 min.)	Use number line models to connect the hourglass video to finding LCM *(SE p. 10)*	1.2.5 Your Turn! (10 min.)	Students find LCM of pairs of numbers and in real-world contexts *(SE p. 13)*
	1.2.3 Try It (10 min.)	Students find LCM of pairs of numbers and in real-world contexts *(SE p. 11)*	1.2.6 Wrap-Up (~5 min.)	Self-assessment rating of confidence in finding the GCF and LCM *(SE p. 14)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: • Multiples • Ellipses • Infinite • Least common multiple (LCM) <u>Additional Terms:</u> N/A	• N/A	• Warm-Up 1.2.1 contains a video to accompany the question prompts.

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may confuse GCF and LCM. • Students may struggle to know when to start/stop their lists or with what to do if they do not see a common multiple. • Students may falsely generalize the LCM as always being the product of the two numbers.	• Consider drawing specific attention to the differences in finding the GCF in Lesson 1 to finding the LCM in this lesson. • Consider adding to the word wall or anchor chart from Lesson 1 by adding new vocabulary words introduced in this lesson, including LCM and multiples. Referencing an anchor chart can help clarify the difference between factors and multiples and minimize confusion.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Consider using the hourglass timing video in 1.2.1 Warm-Up and other real-world contexts throughout the lesson as an opportunity for students to notice and wonder about math in the real world in 1.2.2 Guided Instruction. Consider guiding students through "Notice and Wonder" routines as they ask questions and wonder about how the hourglasses work and connect to finding LCM in 1.2.4 Collaboration.	**MA.K12.MTR.7.1: Real-World Contexts** Students will find the LCM of two numbers within real-world contexts in the 1.2.2 Guided Instruction, 1.2.5 Your Turn!, and 1.2.3 Try It portions of this lesson. Consider highlighting the connection of LCM to everyday experiences. Possible other real-world contexts could include finding when multiples overlap related to time or distance.
Differentiate Instruction	Consider using the problems within the 1.2.3 Try It section as an opportunity to find the LCM using a variety of strategies within student math journals or as an interactive notebook entry. This entry could be used to compare with similar work on the GCF in Lesson 1 and can be a reference for students throughout this unit to help clarify the difference between LCM and GCF. Consider using the number line models from 1.2.2 Guided Instruction section as a support for visual learners in following sections. During 1.2.5 Your Turn! consider having students create their own version of a real-world context problem that includes using the LCM to find when multiples overlap related to time or distance.	
Lesson Notes:		

1.2.1 Warm-Up: Hourglass Timing

Component Overview: *Display the "Hourglass Timing" video for the entire class. The goal of this video is to get students to wonder about the situation presented: three different sand timers set to empty at different rates and wondering when all three timers will run out at the same time. After watching the video, first allow students to write down questions they have as they watch the video or view the images. Allow students to share with a partner, then as a class discuss how each timer is running at a different rate, which will result in a different number of flips during the 30-minute span of time for each timer.*

1.2.1 Warm-Up: Hourglass Timing

The video shows three different hourglass sand timers. The timers are all started at the same time. As one of the times runs out, it is flipped over again to restart.

Several screenshots taken from the video are shown.

1. Write one question you have as you watch the video or review the images.
   Answers will vary. What could we change about the timers, besides the amount of sand to make them empty quicker?

2. Compare questions with your partner.

3. Record your partner's question in the table and have them initial that you recorded it correctly. If you had the same question, work with your partner to generate a different question to ask about the video. *You are the only person who should write in the first column of the table.*

My Partner's Question From the Video	Initials
Answers will vary. What could we change about the timers, besides the amount of sand to make them empty slower?	

Notice and Wonder
After watching the video, consider implementing a Notice and Wonder routine using partner work in this activity. Encourage discourse by facilitating a conversation based on student questions and the video. Ideally, the conversation would lead toward a question that can frame the learning target on LCM, "When will all three timers have no sand in the top half at the same time?" Consider probing questions to guide students toward that framing question.

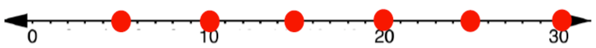
1.2.2 Guided Instruction: Least Common Multiple

Component Overview: *Students explore the concept of finding the LCM with the three hourglass timers from the Warm-Up using a number line model. Be sure to read and emphasize definitions and examples of multiples and the LCM within the section progression. Consider guiding students through this section as a class to discuss different strategies, clarify any misconceptions, and have students write down definitions of multiples and the LCM in a math journal or notebook.*

1.2.2 Guided Instruction: Least Common Multiple

1. Three timers were used in the video, a 2-, 3-, and 5-minute timer.

   a. On each number line model, plot a point indicating the time at which each hourglass would need to be flipped over to restart the timer during a 30 minute period.
   Answer shown in table.

Time	Points Where the Timer Needs to be Restarted
2-minute timer	
3-minute timer	
5-minute timer	

   b. What do you notice about the times plotted on the number line for each timer?
   Answers will vary. I notice the overlap of some of the times. I notice that the 2-minute timer had the most points and needed to be flipped the most.

Multiples of positive whole numbers are the product of that number multiplied by a positive whole number.

For example, the ***multiples*** of 2 are $2, 4, 6, 8 \ldots$.

The **ellipses** (…) at the end of the list of multiples means that the list continues beyond the last number listed. There are an **infinite** number of multiples for any given whole number.

The benchmark for this lesson includes finding the LCM of two numbers, so use questioning prompts to help students identify pairs of timers within the number line model. For example, when answering the question, "What are all the common multiples among the three timers?" include pairs of timers, as well as later looking at all three for added complexity.

Consider asking probing questions such as the following:
- *What do the points on the line represent?*
- *How does this connect to the video we watched in the Warm-Up?*
- *How do these dots relate to one another among all three number lines?*

1.2.2 Guided Instruction: Least Common Multiple (continued)

2. Use the number line models for each of the timers and the definition of multiples to answer the questions.

   a. What are all the common multiples among the three timers?
   30 is the common multiple between all three timers.

   b. What do these common multiples represent in this scenario?
   Answers will vary. When all the timers would be flipped at the same time. When the sand runs out for all the timers simultaneously.

   c. After how many minutes will the 2- and 5-minute timers run out of sand at the same time?
   After $10, 20$, and 30 minutes.

   d. After how many minutes will the 3- and 5-minute timers run out of sand at the same time?
   After 15 and 30 minutes.

The ***Least Common Multiple***, or ***LCM,*** is the lowest number that is a multiple of two or more given numbers.

3. Use the definition to complete the statement.

   The least common multiple (LCM) of 2, 3, and 5 is $\underline{30}$.
   In the context of the hourglass timers, this value represents *when the timers will run out simultaneously*.

4. Find the LCM of 8 and 6.
   LCM $= 24$

Consider keeping the video on display for any of the parts in question 2. Revisiting the video may help visual learners answer the questions if they are able to re-watch important segments or see how the hourglasses flipping relates to the multiples on the number line model.

When modeling finding the LCM of 8 and 6, it is helpful to model using an organized list (vertical or horizontal) and thinking aloud about when to stop the list versus when to keep going. Also consider circling the LCM in both lists. This strategy coupled with the think aloud can serve as an entry point for both visual and auditory learners.

1.2.3 Try It: Least Common Multiple

Component Overview: *Consider having students work in partners for support as they continue practicing using organized lists, number line models, or other strategies to determine the LCM of pairs of numbers and in real-world contexts. Another option is for students to record their work and answers in a math journal or interactive notebook page as a reference to look at in future lessons and to compare to finding the GCF in Lesson 1.*

1.2.3 Try It: Least Common Multiple

1. Find the LCM of each pair of numbers.

   a. 10 and 4
   $LCM = 20$

   b. 5 and 13
   $LCM = 65$

   c. 7 and 21
   $LCM = 21$

2. Every 4 weeks, Fort Caroline Middle has an early release day. Every 5 weeks, they host an assembly with a guest speaker for all students. After how many weeks will Fort Caroline Middle have an assembly and early release in the same week?
   20 *weeks*

3. A radio station is running two ongoing promotions. For the first promotion, every 30^{th} caller wins a pair of movie tickets. For the second promotion, every 100^{th} caller wins an annual pass to a theme park. Which caller number would be eligible for both prizes?
   300

4. Vincenzo and Marie both joined a local gym. Vincenzo plans to go to the gym every 2 days at 7:00 am when they open. Marie plans to go every 3 days at 7:00 am. If they both joined the gym on the same day, after how many days will they arrive at the gym on the same day?
   6 *days*

Help students analyze their responses in problem 1 and recognize that the LCM is not always the product of the two numbers. After students finish all three parts of question 1, ask them, "What do you notice about the LCMs of the first three pairs of numbers?" Be sure to point out that sometimes the LCM is the product of the two factors; sometimes one of the numbers itself is the LCM; and sometimes the LCM is larger than both numbers, but less than the product.

What are the multiples of __ and __? What are the common multiples of __ and __? Which is the least common multiple? How do I know when I have found enough for finding the LCM?

1.2.4 Collaboration: Least Common Multiple

Component Overview: *Students work in partners to use the digits 1 through 9 one time each to make a statement about two factors and the LCM true. Consider setting a time limit for this portion of the lesson, as there are many correct answers and finding possibilities may take students more time than is available. If time is limited, this can also be used on a different day as a review or open-ended task.*

1.2.4 Collaboration: Least Common Multiple

1. Place a digit in each blue box to make a true statement. Use only digits 0 to 9, at most one time each.

   Factor A: ☐☐

   LCM: ☐☐☐

   Factor B: ☐☐

   Answers will vary. For example, 28 *and* 35: *LCM* $= 140$

2. What were some of the challenges you faced with this problem?
   Answers will vary. Some of the LCMs were large numbers so finding them took a long time. It was challenging to not reuse digits.

3. What methods did you use to solve this problem?
   Answers will vary. We wrote out all the multiples until we found a match.

Consider using probing questions while monitoring student collaboration for students who may need additional support:

- *What do you notice about the number of boxes available?*
- *What do you notice about the LCM?*
- *What must be true for the factors and the LCM?*
- *What do you think is a good strategy to get started?*
- *Did you use a number more than once in any of the boxes?*

This task can be complex because there are many correct answers, and the LCMs can be large values. To support learners, help students think aloud and verbalize what they notice before getting started, including recognizing that the factors are two digits and can include 0 as the first digit to represent a value with only ones. For example, 07 can be used to represent the factor of 7.

1.2.5 Your Turn!

Component Overview: *Students individually apply strategies to finding the LCM for a pair of numbers and real-world problems. Have students complete this work independently as you circulate to check for student understanding.*

1.2.5 Your Turn!

1. Find the LCM of each pair of numbers.

   a. 15 and 20
   LCM $= 60$

   b. 18 and 24
   LCM $= 72$

   c. 22 and 11
   LCM $= 22$

2. Jasmine and Teia are running a race. Jasmine runs at a rate of 4 minutes per lap and Teia runs at a rate of 6 minutes per lap. If they both start running at the same time, after how many minutes will they both be at the starting line at the same time again?
   12 *minutes*

3. Lin's uncle is opening a bakery. On the bakery's grand opening day, he plans to give away prizes to the first 50 customers that enter the shop. Every 9^{th} customer will get a free blueberry muffin. Every 12^{th} customer will get a free slice of carrot cake. Which customer will get both free bakery items?
   36th *customer*

How do you know when you have found enough multiples to determine the LCM? How can you determine the LCM?

Consider encouraging students to use visual models to find the LCM for the word problems in this section. For example, students can use a number line model like in the Guided Instruction example. Students may also use organized lists and skip count to determine multiples, then circle the LCM as a visual strategy.

Consider extending student understanding of the LCM by asking students to think of other contexts, besides those in this section, where the LCM is used to solve a real-world problem, including when multiples overlap related to time, distance, or order.

1.2.6 Wrap-Up: Reflection

Component Overview: *Use this self-evaluation as an opportunity for students to reflect on their level of mathematical confidence in understanding GCF and LCM from Lessons 1-2.*

1.2.6 Wrap-Up: Reflection

1. Rate your level of mathematical confidence on the concepts listed by circling the statement that best describes your understanding.
   Answers will vary.

- **Greatest Common Factor (GCF)**

I don't know what GCF is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.

- **Least Common Multiple (LCM)**

I don't know what LCM is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.

If time permits, consider asking students to explain the difference between the GCF and LCM with a partner before completing this self-evaluation. This practice of recalling the difference both supports conceptual understanding and provides an entry point for auditory learners in the self-evaluation.

For students who may finish early, consider having them provide examples of GFC and LCM to support their statements.